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Research article

EEG correlates of math anxiety during arithmetic problem solving: Implication for attention deficits

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ABSTRACT

Anxiety about math can lead to long-term negative consequences related to academic achievement and professional success. However, it remains unclear how elevated math-anxiety modulates brain activity while solving arithmetic problems. In the current study, we recorded electrophysiological responses throughout arithmetic problem solving, both at the period of anticipating an upcoming arithmetic problem and solving an arithmetic problem. Results showed that, after controlling for mathematical performance, people with higher math anxiety tended to show stronger beta band oscillation and P300 amplitude while expecting the arithmetic problems, as well as stronger gamma band activity while solving the arithmetic problems. These results suggest that individuals highly anxious about math might use more attentional resources during the course of anticipating the upcoming arithmetic problems, and showed greater attentional bias toward arithmetic problems during solving arithmetic problems.

1. Introduction

Mathematical anxiety is defined as a negative emotional reaction especially related to mathematical processing [1]. According to the Attention Control Theory (ACT), individuals with high anxiety level might spend more cognitive resources trying to suppress negative emotional responses, leading to increased stimulus-driven attention, increased top-down attention [2]. Several psychophysiological measures are associated with attention deficits: N100 and P200 components are thought to index early sensory processing and low-level attention allocation [3,4]. P300 is thought to reflect the stimulus evaluation time and attentional requirements needed to ultimately repress the information [5]. Brain oscillations can also provide an effective means to control the timing of neuronal firing [6]. Beta oscillations are related to behavioral inhibition [7,8]. Gamma oscillations are related to visual attention processing [9]. However, the empirical evidence for the attention deficit of high math anxious individuals is still lacking.

Studies have demonstrated neural differences between those with high math anxiety (HMAs) and those with low math anxiety (LMAs) during mental calculation [10–12]. The amplitude of P200 component was found to be larger in HMAs than their LMA peers during arithmetic problem solving [11]. This was inconsistent with another finding showing that HMAs exhibited reduced positive ERP amplitude at fronto-central locations around 200 ms after stimulus onset during arithmetic problem solving [10]. A core problem with the existing research is that the neuropsychological mechanism might be confounded with math performance [13]. Behavioral studies have shown that compared with those who have low levels of math anxiety, individuals who are highly anxious perform poorly on multiple types of mathematical operations [14,15].

In the current work, we sought to characterize the neural differences between HMAs and LMAs after controlling for their behavioral performance. Participants were asked to perform the arithmetic verification task while recording their EEGs. We extracted and analyzed the event-related potentials (ERPs) and event-related synchronization/desynchronization (ERS/ERD) of both at the period of anticipating the arithmetic problems and the period of solving the problems. According to the ACT, we assumed that HMAs exhibit increased stimulus-driven attention and increased goal-directed attentional control. Specifically, after statistically controlling for behavioral performance, we expected

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Table 1
Means and standard errors for age, sex, math anxiety, trait anxiety, working memory, and IQ for the LMA and HMA groups.

	Age	Gender	MA	TA	VWM	IQ
HMA	19.88(1.45)	17/21	90.52(9.48)	43.82(5.19)	8.79(1.96)	5.10(1.11)
LMA	19.53(1.5)	18/19	44.00(6.24)	45.56(5.64)	8.11(2.56)	5.25(1.08)
t	–	–	24.95	1.38	1.28	0.57
p	–	–	< .001	0.17	0.20	0.57

Note: MA, mean scores on the Chinese brief version of the Mathematics Anxiety Rating Scale; TA, mean scores on Spielberger's State and Trait Anxiety Inventory; VWM, mean stanine scores for verbal working memory test; IQ, mean stanine scores on the non-verbal Matrix reasoning test.

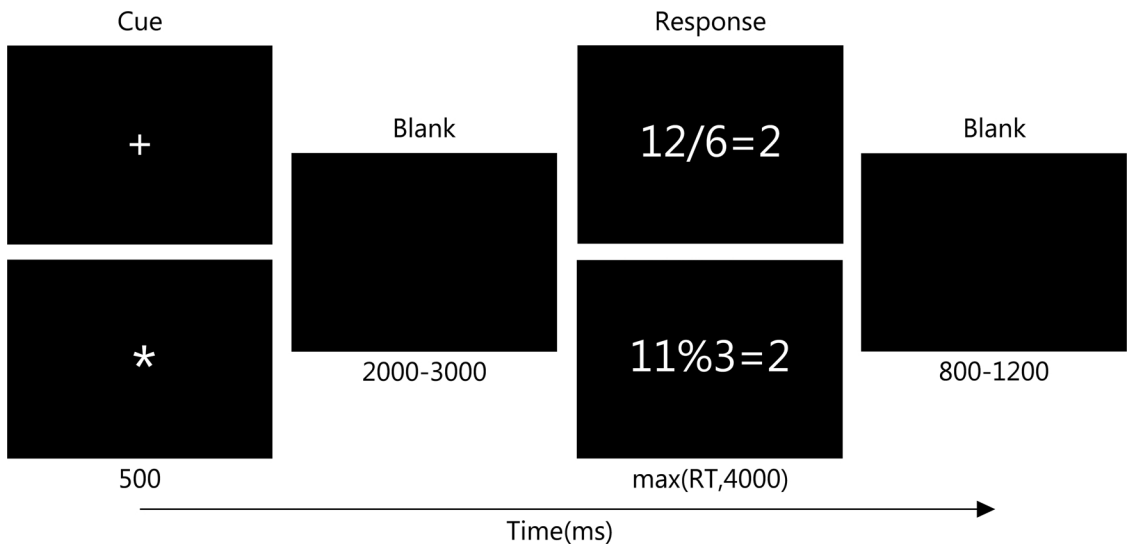

Fig. 1. Procedure of the arithmetic task during EEG recording. A fixation sign "+" or "*" was presented in the center of the screen for 500 ms to cue the beginning of the trial. Fixation was followed by a 2–3 s jittered blank screen. The arithmetic problems appeared at the center of the screen until the participants responded. An 800–1200 ms blank was presented following the response.

to observe greater N100/P200/P300 amplitude, as well as greater activity in beta/gamma-band oscillation in HMAs than in LMAs.

2. Methods

2.1. Participants

Participants were 80 right-handed college students. Math anxiety groups were determined using the Chinese brief version of the Mathematics Anxiety Rating Scale (SMARS) [16]. Among the pre-screened participants (n = 253), 40 participants who scored above 80th percentile were included in the HMA group and 40 participants who scored below 20th percentile were included in the LMA group. Two participants were excluded due to technical problems during data collection, and four were excluded due to low EEG data quality during data acquisition, leaving 38 participants in HMA and 36 in LMA. Detailed demographic information is shown in Table 1. We then assessed the general cognitive ability of these participants, including IQ and verbal working memory. Trait anxiety was assessed using the State-Trait Anxiety Inventory (STAI) [17]. Groups differed in math anxiety ($t(72) = 25.23$, $p < 0.001$), but not in trait anxiety($t(72) = 1.38$, $p = 0.17$) and state anxiety ($t(72) = 0.46$, $p = 0.65$). The study was approved by the Medical Ethical Committee of Medical School in Shenzhen University, Shenzhen, China.

2.2. Materials

2.2.1. Controlling variables

The non-verbal matrix reasoning test was designed based on Raven's Standard Progressive Matrices [18], which is used to assess general intelligence. Participants were asked to identify the missing section of an image according to the inherent rules of the picture. The verbal working-memory task was adapted from the digit-span task from the Wechsler Intelligence Scale [19]. Participants were asked to remember the order of the digits and key them into the computer at the end of each series. Both forward and backward tasks were used and the overall performance was calculated.

2.2.2. Arithmetic task during the EEG recording

Two types of arithmetic problems were included. One was simple division problems (e.g., 12 ÷ 3 = 4) and the other was mod problems (e.g., 11%3 = 2). The dividends of both operations were single digits. Mod problem is a type of division problem for which the solution is the remainder. For example, the solution for 11%3 is 2. For both types of math problem, the participants needed judge whether the presented solution was correct. Judgments were made by pressing the indexed keys. There were 56 trials for each type of problem. Participants were trained and given practice trials until they fully understand both operations.

2.3. Procedure

The questionnaires and cognitive assessments were carried out two weeks before the EEG recording. During EEG recording, stimuli were presented visually at the center of the computer screen. The mean visual angle was 2.43° for the division and 2.45° for mod problems. Division and mod problems were presented in separate mini blocks (4 trials in each block). The detailed trial procedure was shown in Fig. 1. Participants were asked to keep their head steady and avoid eye blinking. The presentation of stimuli and the collection of behavioral data were controlled by E-Prime 2.0 (Psychology Software Tools Inc., Sharpsburg, PA, USA).

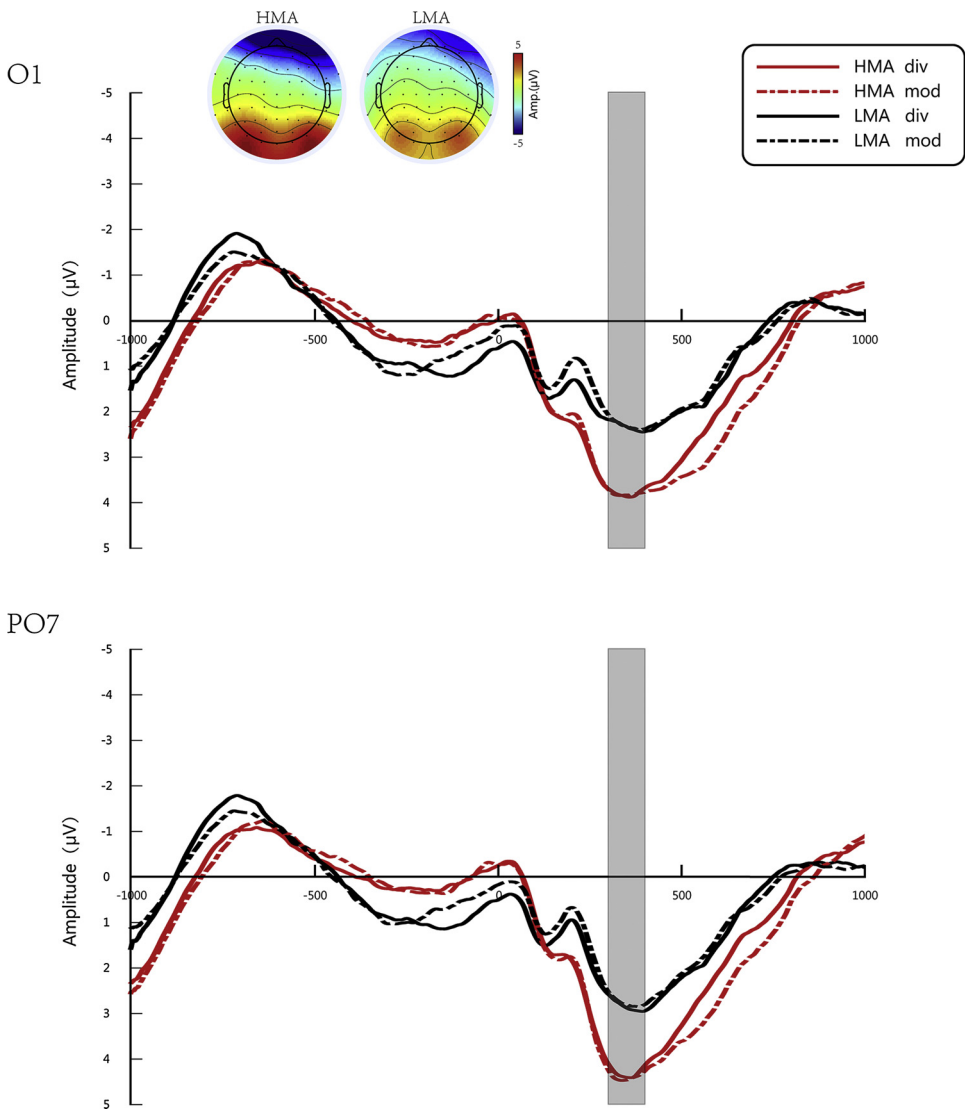

Fig. 2. P300 of the ERP in the HMA (red line) and LMA (black line) groups at O1 and PO7 and the corresponding spatial distribution of the effect size over all electrodes at the scalp surface in both groups in cue-induced ERP.

2.4. Electrophysiological recording

EEG data were collected with Brain Products from 64 Ag–AgCl scalp electrodes placed according to the extended International 10–20 system. EEG channels were continuously digitized at a rate of 500 Hz and band-pass filtered at 0.01–100 Hz. EEG activity was recorded referentially against FCz and off-line rereferenced to the global average reference. Impedance for all electrodes was less than 10 kΩ during the task. Electrooculographic (EOG) signals were recorded using surface electrodes to monitor ocular movements and eye blinks.

2.5. Data processing

2.5.1. Behavioral data analysis

For reaction times (RTs), incorrect responses were removed from the analysis. RTs that exceed three times the standard deviation were removed as outliers. Mean RT and accuracy were compared using a two-way mixed ANOVA, with “experimental groups” as between-subject factor and “operation type” as a within-subject factor.

2.5.2. EEG data preprocessing

EEG data were processed using EEGLAB [20]. Continuous EEG data were band-pass filtered between 1 and 45 Hz. Two EEG epochs were extracted using: 1) a 2000 ms epoch ranging from -1000 ms before the cue onset until 1000 ms after the cue onset; 2) a 2000 ms epoch ranging from -1000 ms before the stimulus onset until 1000 ms after the stimulus onset. Artifact rejection was implemented with the “pop_autorej ()” function in EEGLAB, to detect and delete extremely large potential fluctuations. Remaining trials contaminated by eye-blinks and movements were corrected using an Independent Component Analysis (ICA) algorithm. EEG epochs were visually inspected. Mean valid trials for division problems were 54 (HMA) and 54 (LMA) and those for mod problems were 55 (HMA) and 55 (LMA) in the cue-induced analysis. Epochs of correct responses were analyzed for the stimulus-induced epochs. Mean valid trials for division problems were 49 (HMA) and 51 (LMA) and those for mod problems were 41 (HMA) and 45 (LMA) in the stimulus-induced analysis.

2.5.3. Time-domain analysis

Three components were examined: N100, P200 and P300. The mean amplitudes were compared using a three-way mixed ANOVA, with “experimental groups” as between-subject factor, “operation type” and “electrode” as within-subject factors.

Cue-induced ERP. Because we did not observe any negative

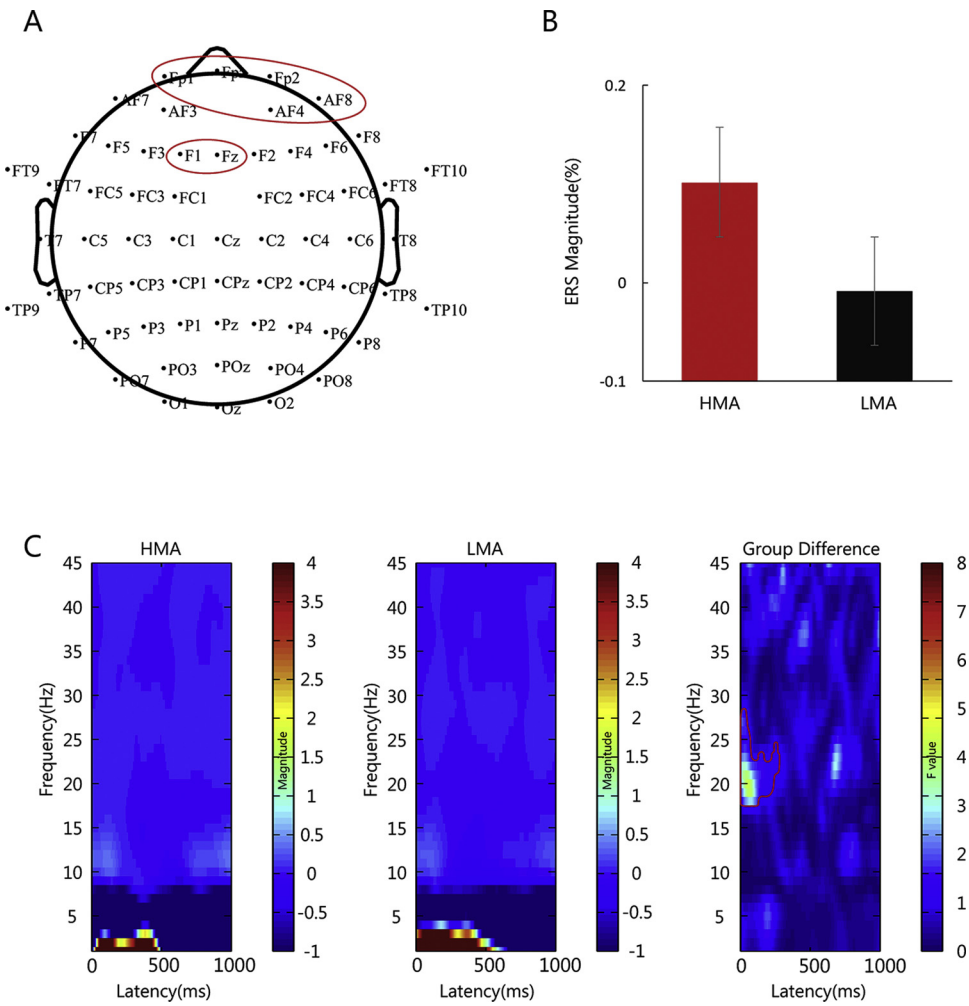

Fig. 3. ROI (Fpz, Fp1, Fp2, AF4, AF8, F1 and Fz, 18–26 Hz, 1–312 ms) in time-frequency analysis of cue-induced EEG response. A. The scalp distributions of the ROI; B. The mean power for the two types of operation within the ROI for both HMA and LMA respectively, error bars represent the SEM; C. The left two panels, time-frequency representations of mean EEG responses of two types of operations for HMA and LMA, the color scale represents the average increase (ERS%) or decrease (ERD%) of oscillation magnitude, relative to a prestimulus reference interval (0.9–0.1 s before the onset of the cue), for HMA and LMA respectively. x-axis, time (ms); y-axis, frequency (Hz). The right panel, the main effect of the group factor on the time-frequency representations within the, in two-way mixed ANOVA, The color scale represents the F-values for the main effect of group. HMAs showed significant greater magnitude than LMAs within this ROI in the hierarchical linear regression analysis ($\triangle R^2 = 0.15$, Bonferroni Corrected $p = 0.028$).

deflection around 100 ms of the waveform, N100 was not analyzed in the cue-induced ERPs. P200 was defined as the most positive deflection between 140–160 ms and located at the occipital electrodes (PO7, PO8, O1, and O2). P300 was defined as the most positive deflection between 250–350 ms located at the occipital electrodes (PO7, PO8, O1, and O2) (Fig. 2).

Stimulus-induced ERP. N100 was defined as the most negative deflection between 60–80 ms located at the occipital electrodes (Pz, Oz, CPz, POz, P1, and P2). P200 was defined as the most positive deflection between 190–210 ms and located at the frontal electrodes (FC1, FC3, C1, C2, C3, CZ, FC2, F1 and F3). P300 was defined as the most positive deflection between 300–400 ms located at the occipital electrodes (P1, P2, P3, P4, P6, Pz, PO8, PO3, PO4, and POz).

2.5.4. Time-frequency-domain analysis

For each participant, EEG signals from all trials were transformed to the time-frequency domain using a windowed Fast Fourier with fixed 250-ms sliding Hanning window. As magnitude differences between HMA and LMA have been observed at a latency of 200 ms after stimulus onset [10,11], a 250 ms Hanning window provides a sufficiently high frequency resolution of 4 Hz, and is suitable for identifying these responses for it [21]. For each time course, the windowed Fourier transform yielded a complex time-frequency estimate at each point of the time-frequency plane, extending from −1000 to 1000 ms (in 2 ms intervals) in the time domain, and from 1 to 45 Hz (in 1 Hz intervals) in the frequency domain. We chose 1–45 Hz because we ought to include both lower band (1–30 Hz) and higher band (e.g., gamma band, 30 Hz and above) in our analysis. The spectrogram was baseline-corrected (reference interval: −900 to −100 ms) at each frequency according to the formula: $ER(t,f) = \frac{F(t,f) - R(f)}{R(f)}$, where $F(t,f)$ is the signal power at a given time t and at a given frequency f, and $R(f)$ is the signal power of the frequency f averaged within the reference interval. The wavelet transform was computed on the average of the single trials.

We adopted a data-driven approach to find out the time-frequency point that showed significant different value between these two experimental groups. Point-by-point two-way mixed ANOVA, combined with nonparametric permutation testing [22], was used to assess the differences between experimental groups on modulations of EEG oscillation power.

The electrodes with the maximal group difference signal were selected to compute mean values for each significant time-frequency region of interest (ROI) of each participant. We first computed the mean value of all time-frequency points within that ROI for each participant for each experimental condition. The mean values were then compared using a two-way mixed ANOVA, with “experimental groups” as between-subject factor and “operation type” as within-subject factor.

In order to rule out the impact of performance differences, as well as other confounding factors between the HMA and LMA groups, a hierarchical linear regression model was used to directly test the group effect of math anxiety. The behavioral performance was included in the hierarchical linear regression model. Other irrelevant variables such as sex, age, trait anxiety, IQ, and verbal working memory were also included in the hierarchical model. We calculated the mean amplitude of both operation types for the selected electrodes in the component that show significant group effect in the mixed-ANOVA analysis, and considered the mean value as the dependent variable (Y). The linear

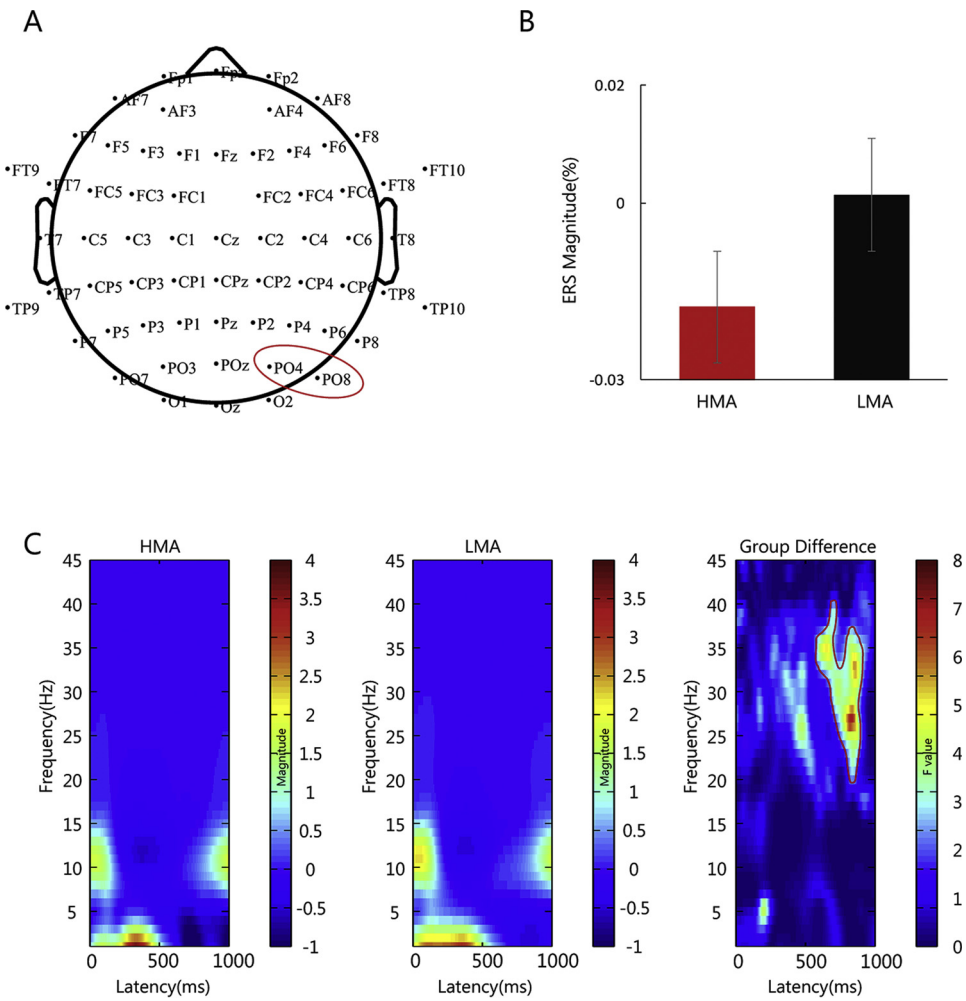

Fig. 4. ROI (PO4 and PO8, 31–40 Hz, 590–894 ms) in time-frequency analysis of cue-induced EEG response. A. The scalp distributions of the ROI; B. The mean power for the two types of operation within the ROI for both HMA and LMA respectively, error bars represent the SEM; C. The left two panels, time-frequency representations of mean EEG responses of two types of operations for HMA and LMA, the color scale represents the average increase (ERS%) or decrease (ERD%) of oscillation magnitude, relative to a prestimulus reference interval (0.9–0.1 s before the onset of the cue), for HMA and LMA respectively. x-axis, time (ms); y-axis, frequency (Hz). The right panel, the main effect of the group factor on the time-frequency representations within the, in two-way mixed ANOVA, The color scale represents the F-values for the main effect of group. HMAs showed significantly smaller magnitude than LMAs within this ROI in the hierarchical linear regression analysis ($\triangle R^2 = 0.11$, Bonferroni Corrected $p = 0.020$).

regression model equation can be expressed as:

$$Y = \beta 0 + \beta 1 \bullet X_{ACC} + \beta 2 \bullet X_{RT} + \beta 3 \bullet X_{age} + \beta 4 \bullet X_{sex} + \beta 5 \bullet X_{IQ} + \beta 6 \bullet X_{VWM} + \beta 7 \bullet X_{TA} + \beta 8 \bullet X_{group}, \quad (1)$$

where ACC = Accuracy, RT = Reaction Time, IQ = Intelligent quotient; VWM = Verbal working memory, and TA = Trait Anxiety.

3. Results

3.1. Behavioral results

For RT, we found no significant interaction between the two main factor ($F_{(1,\ 72)} = 3.74$, $p = 0.06$, HMA > LMA), as well as no significant Group main effect ($F_{(1,\ 72)} = 0.61$, $p = 0.44$). For accuracy, we found no significant interaction between the two ($F_{(1,\ 72)} = 2.83$, $p = 0.10$), but significant main effects for the Experimental Group (HMA < LMA, $F_{(1,\ 72)} = 8.70$, $p = 0.004$, $\eta_p^2 = 0.11$).

3.2. Time-domain results

3.2.1. Cue-induced ERP

For P200 (140–160 ms, at PO7, PO8, O1, and O2), we found significant interaction effect between the operation type and the experimental group ($F_{(1,\ 72)} = 6.99$, $p = 0.01$, $\eta_p^2 = 0.09$), as well as significant group main effect (HMA > LMA, $F_{(1,\ 72)} = 4.47$, $p = 0.04$, $\eta_p^2 = 0.06$) at the PO7 and O1 electrode. Post-hoc analysis showed greater P200 amplitude in HMA group than that in LMA group for mod problem (PO7: $t(72) = 1.90$, $p = 0.009$; O1: $t(72) = 1.56$, $p = 0.005$), but not for the division problems ($ps > 0.05$). We observed neither significant interaction effect between "operation type" and "experimental group" nor significant group main effect ($ps > 0.05$) at the right hemisphere (PO8 and O2). Hierarchical linear regression analysis suggested that the group difference between HMAs and LMAs (Bonferroni Corrected p > 0.05) could no longer exist after regressed out the confounding factors, such as the trait anxiety, IQ, behavioral performance, verbal working memory ability, as well as their gender. For P300 (250–350 ms, at PO7, PO8, O1, and O2), we found no significant interaction effect between the operation type and the experimental group ($p > 0.05$). We found significant group main effect (HMA > LMA, $F_{(1,\ 72)} = 5.95$, $p = 0.02$, $\eta_p^2 = 0.08$) at the PO7 and O1 electrode. Hierarchical linear regression analysis suggested that HMAs elicited significant greater P300 amplitude than LMAs ($\triangle R^2 = 0.08$, Bonferroni Corrected p < 0.05) after regressed out the confounding factors. We observed neither significant interaction effect between "operation type" and "experimental group" nor significant group main effect ($ps > 0.05$) at the right hemisphere electrodes (PO8 and O2).

3.2.2. Stimulus-induced ERP

For N100, P200 and P300, we observed neither significant interaction effect between group and operation type ($ps > 0.05$) nor significant group main effect ($ps > 0.05$) in the time period of solving the arithmetic problems.

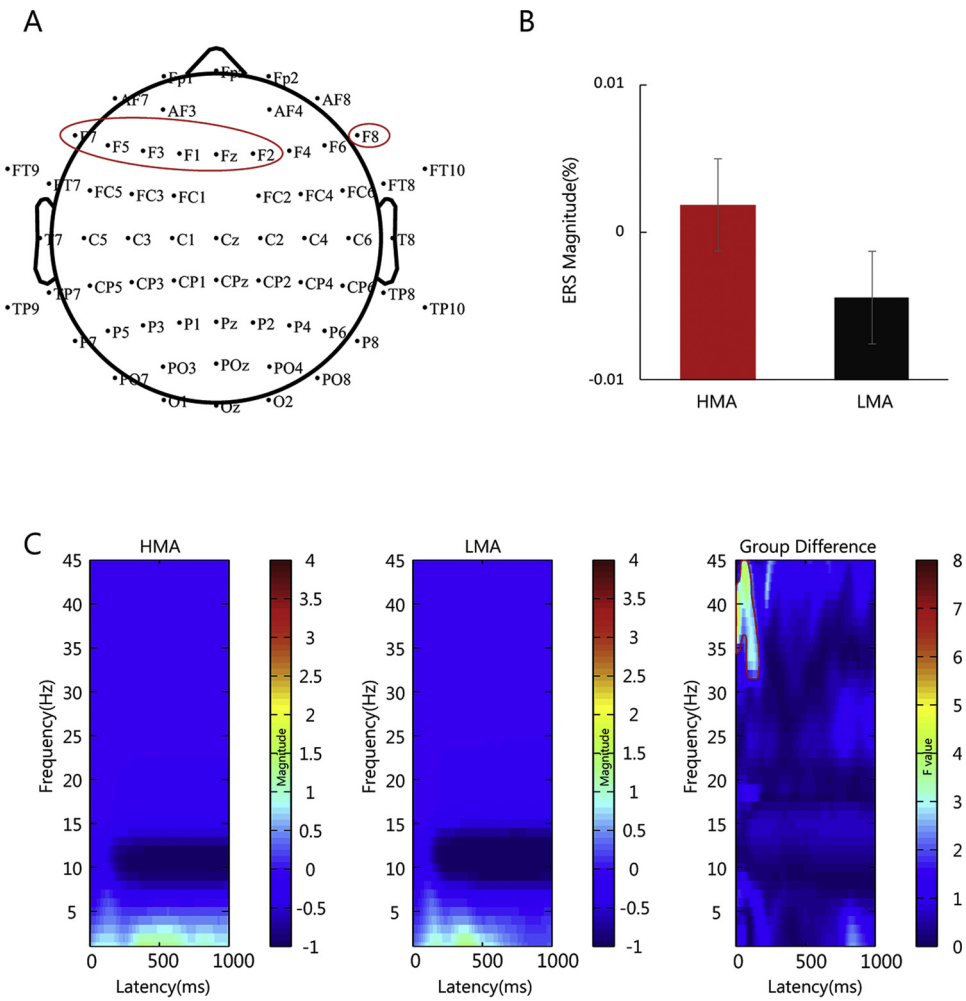

Fig. 5. ROI (P1, P2, P3, P5, Pz, P7, and P8, 32–45 Hz, 1–202 ms) in time-frequency analysis of stimulus-induced EEG response. A. The scalp distributions of the ROI; B. The mean power for the two types of operation within the ROI for both HMA and LMA respectively, error bars represent the SEM; C. The left two panels, time-frequency representations of mean EEG responses of two types of operations for HMA and LMA, the color scale represents the average increase (ERS%) or decrease (ERD%) of oscillation magnitude, relative to a prestimulus reference interval (0.9–0.1 s before the onset of the cue), for HMA and LMA respectively. x-axis, time (ms); y-axis, frequency (Hz). The right panel, the main effect of the group factor on the time-frequency representations within the, in two-way mixed ANOVA, The color scale represents the F-values for the main effect of group. HMAs showed significant greater magnitude than LMAs within this ROI in the hierarchical linear regression analysis ($\triangle R^2 = 0.09$, Bonferroni Corrected $p = 0.004$).

3.3. Time-frequency domain results

Two significant clusters (one above 30 Hz and one below 30 Hz) were identified in the point-to-point non-parametric permutation test for cue-induced and stimulus-induced EEG respectively, which yield four ROIs for the subsequent ROI analysis.

Figs. 3 and 4 showed the time-frequency distributions classified in both groups, as well as the statistical value of between groups for the cue-induced responses. The magnitude of the cue-induced responses was significantly modulated by the experimental group at 2 ROIs: The first ROI included Fpz, Fp1, Fp2, AF4, AF8, F1 and Fz (18–26 Hz, 1–312 ms). The interaction between group and operation type was not significant ($p > 0.05$), the group main effect was significant ($F_{(1, 72)} = 9.99$, $p = 0.002$, $\eta_p^2 = 0.12$). Furthermore, HMAs showed significant greater magnitude than LMAs within this ROI in the regression analysis ($\triangle R^2 = 0.15$, Bonferroni Corrected $p = 0.028$). The second ROI included PO4 and PO8 (31–40 Hz, 590–894 ms). The interaction between group and operation type was not significant ($p > 0.05$), the group main effect was significant ($F_{(1, 72)} = 8.17$, $p = 0.006$, $\eta_p^2 = 0.10$). HMAs showed significantly smaller magnitude than LMAs within this ROI in the regression analysis ($\triangle R^2 = 0.11$, Bonferroni Corrected $p = 0.020$).

Fig. 5 shows the time-frequency distributions classified in both groups, as well as the statistical value of between groups for the stimulus-induced responses. The magnitude of the cue-induced responses was modulated by the experimental group at 2 ROIs: The first ROI included P1, P2, P3, P5, Pz, P7 and P8 (32–45 Hz, 1–202 ms). The interaction between group and operation type was not significant ($ps > 0.05$), the group main effect was significant ($F_{(1, 72)} = 14.42$, $p < 0.001$, $\eta_p^2 = 0.17$). HMAs showed significant greater magnitude than LMAs within this ROI in the regression analysis ($\triangle R^2 = 0.09$, Bonferroni Corrected $p = 0.004$). The second ROI included F1, F3, F5, F7, and AF7 (1–8 Hz, 1–106 ms). The interaction between group and operation type was not significant ($ps > 0.05$), the group main effect was significant ($F_{(1, 72)} = 7.20$, $p = 0.009$, $\eta_p^2 = 0.91$). HMAs did not show any significantly greater magnitude than LMAs within this ROI in the regression analysis (Bonferroni Corrected $p > 0.05$).

4. Discussion

Here, we characterized the brain electrophysiological activity across the entire time course of arithmetic problem solving in individuals who have high and low math anxiety. The main findings could be summarized as follows: Irrespective of participants' behavioral performance or other factors such as sex, age, trait anxiety scores, IQ, or verbal working memory, HMAs exhibited significant greater P300 amplitude, greater beta-band power oscillation, as well as smaller gamma band activity than LMAs while anticipating the arithmetic problems, and exhibited greater gamma band activity while solving the arithmetic problems than LMAs. This finding provides empirical evidence for the attentional deficits in high math-anxious individuals, and is in line with recent findings showing diffused and unstructured network related to cognitive related processing for HMAs [12,23].

Our finding related to the P200 might give help to explain the previous inconsistent findings. The amplitude of P200 was found to be larger in HMAs than in their LMA peers during multi-digit arithmetic

problem solving [11]. Klados et al. [10] showed an opposite result that HMAs exhibited reduced P200 amplitude around fronto-central locations during arithmetic problem solving task. In our preliminary mixed ANOVA analysis, the group effect was significant for the P200 amplitude (HMA > LMA). Then this group effect disappeared after the confounding factors were regressed out in the hierarchical linear regression analysis. Thus, the inconsistent results in the previous studies might be due to the confounding factors.

The present finding showed enhanced P300 amplitude for HMA at the posterior sensors, which is line with the literature that HMAs showed an enhanced P3b around the parietal cortex when giving large-split solutions for simple arithmetic problems [23]. P300 amplitude is associated with increased resource deployment [24], which is often larger for emotional than for neutral stimuli [25,26]. The enhanced P300 amplitude for HMAs indicated that HMAs have difficulties in top-down attention inhibition.

The higher beta-band oscillatory activity at the frontal sensors at the period of anticipating the arithmetic problems for HMA could serve as complementary evidence to previous fMRI results that showed HMAs exhibited increased blood oxygenation level dependent signal responses before the onset of arithmetic tasks in the insula and inferior frontal junction [27,28]. High beta-band activity is hypothesized to play an important role in enhancing the neural representation of sensory input in stimulus-driven attention processing, as well as the top-down attention control [29,30]. HMAs might spend more attentional resources while expecting the arithmetic problems, and performance in HMAs compromised because successful shifting of attention and inhibitory processes are limited. Our finding supports the ACT which proposes that HMAs have deficits in the top-down goal-directed attentional system [2].

Gamma-band activity is proposed to integrates neural activities related to a specific sensory object into a stable, salient and coherent representation, which would then be available for further maintenance in short-term memory or encoding into long-term memory [9]. Several relevant results manifested that individuals with high-level anxiety showed increased gamma-band activity, which might be due to attention bias to negative emotional stimuli [31,32]. In accordance with the pattern of general anxiety, our early gamma band activity suggested that visually presented arithmetic problems elicited greater gamma-band activity in HMAs than LMAs in posterior electrodes, suggesting there might exist greater attentional bias for negative stimuli – arithmetic problems – for HMAs.

One limitation of the current study is that we did not counterbalanced the response hand across participants, which might leading the main results be interfered with the motor responses. This may be particularly relevant in the beta frequency at centrolateral electrodes [6]. In conclusion, we provide insight on the whole temporal course of brain psychophysiological activity when individuals with high math anxiety solve arithmetic problems. We also provided evidence for attention deficit in HMA and perhaps EEG markers for objectively diagnosing math anxiety.

Acknowledgments

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